

Estimation and mapping of forest stand density, volume, and cover type using the k -nearest neighbors method

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Abstract

Mapping forest variables and associated characteristics is fundamental for forest planning and management. Considerable effort has been made in Northern Europe to develop techniques for wall-to-wall mapping of forest variables. Following that work, we describe the k -nearest neighbors (kNN) method for improving estimation and to produce wall-to-wall basal area, volume, and cover type maps, in the context of the USDA Forest Service's Forest Inventory and Analysis (FIA) monitoring system. Several variations within the kNN were tested, including: distance metric, weighting function, feature weighting parameters, and number of neighbors. Specific procedures to incorporate ancillary information and image enhancement techniques were also tested. Using the nearest neighbor ($k=1$), Euclidean distance, a three date 18-band composite image, and feature weighting parameters, maps were constructed for basal area, volume, and cover type. The empirical, bootstrap based, 95% confidence interval for the basal area root mean square error (MSE) is (8.21, 9.02) m²/ha and for volume (48.68, 54.58) m³/ha. For the 13 FIA forest cover type classes, results indicated useful map accuracy and the choice of $k=1$ retained the full range of forest types present in the region. The 95% confidence interval, obtained using the bootstrap 0.632+ technique, for the overall accuracy (OA) in the 13 cover type classification was (0.4952, 0.5459). Recommendations for applying the kNN method for mapping and regional estimation are provided. © 2001 Elsevier Science Inc. All rights reserved.

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1. Introduction

The Forest Inventory and Analysis (FIA) program conducted by the USDA Forest Service surveys the United States forest resources on a state-by-state basis. Ultimately, the FIA provides information on forest and related resources for the entire nation. The main goal of FIA is to provide timely information for the development of policies and programs for protection, management, and utilization. This information constitutes the quantitative basis for making sound management, conservation, and stewardship decisions affecting these valuable resources (Anonymous, 1992).

The FIA system is by no means static; it has evolved and adapted in its history of more than 60 years. Birdsey and Schreuder (1992) described how the survey has adapted to

match changing information needs and advances in forest inventory technology. Moreover, intensifying interest in forests and the environment have induced major changes in forest monitoring systems in the last few years. The most significant change in the FIA program in several decades is the recent shift from periodic resurveys to an annual system of field data collection and analysis (Czaplewski, 1995; McRoberts, 1999; Reams, 1999). The report of the second Blue Ribbon Panel on FIA considered the annual inventory a key to the timely collection and analysis of forest inventory data (Anonymous, 1998a). Thus, the USDA Forest Service is seeking to provide more timely and accurate local estimations, broad subject matter coverage, and increased information availability.

Mapping forest variables and associated characteristics is fundamental for forest management. Zhu and Evans (1994) produced one of the few examples of what one might call an FIA map. This forest type classification map was developed for the entire country based on classification of NOAA AVHRR data from different FIA units, however, its coarse

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resolution limits its usefulness. Another example of generalizing FIA land cover classification at a state level with AVHRR data is given by Teuber (1990). FIA is not required to provide for such mapping at a local level. Instead, forest, county, and regional level maps have been developed by a range of federal, state, and private organizations. Most of these maps have been developed for regional forest cover type classification with little or no use of FIA data.

The utility of FIA data would be greatly increased if there were a simple way to use it to develop locally useful maps of a range of variables. For example, maps of forest variables such as basal area, volume, and cover type could be very useful to FIA clientele. In turn, such mapping could greatly improve the precision and accuracy of forest estimates at county and more local levels, and thereby provide essential data for forest management planning.

Considerable effort has been made in Northern Europe to develop techniques for wall-to-wall mapping of forest variables. Tokola, Pitkänen, Partinen, and Muinonen (1996) and Tomppo (1991) applied the *k*-nearest neighbors (kNN) method to produce localized estimates and maps from the national forest inventory data of Finland. Tomppo also incorporated the method into the national forest inventory of Finland on an operational basis. The method shows great promise for the mapping of continuous variables, such as basal area and volume, and for cover type. The method could be easily integrated within existing forest monitoring systems procedures. Yet, there is an important difference between the kNN and traditional classification and estimation techniques. The kNN method is a form of poststratification constrained to the range of plot values of the inventory. In effect, after field plots are taken, they comprise strata with associated variable values. These values are then assigned to the remaining nonselected plot locations according to the similarity of certain features among the sampled and nonsampled plots. As an example, a mature pine plot and its variable values are distributed (assigned) across the landscape to nonsampled locations that are determined to be similar in some sense. Conversely, traditional classification attempts to establish strata according to the inventory plots they may contain. Thus, the kNN retains the full set of inventory specifications and values, while traditional classification typically does not.

The overall objective of this paper is to describe methodology for using the kNN and related techniques, to improve estimation, and to produce wall-to-wall basal area, volume, and cover type maps, in the context of the FIA forest monitoring system. Study details beyond those provided in this paper are available in Franco-Lopez (1999).

2. Background

2.1. kNN estimation procedure

The kNN method is used here to generalize information from field plots to pixels for map production and local area

estimation. The method assumes that similar forest exists within a large reference area covered by a satellite image and that the spectral radiometric responses of the pixels are only dependent on the state of the forest. Several examples can be found in the literature, including: Fazakas and Nilsson (1996), Muinonen and Tokola (1990), Nilsson (1997), and Tomppo (1991, 1993, 1997a, 1997b). We note that Tomppo has led efforts to incorporate the method in forest inventories.

A general description of the kNN method is as follows. The spectral distance, $d_{p,p}$ is computed in the feature space from the pixel p to be classified to each pixel p_i for which the ground measurement or class is known. For each pixel p , take *k*-nearest field plot pixels (in the feature space) and denote the distances from the pixel p to the nearest field plot pixels by $d_{p,p}, \dots, d_{p,p}$ ($d_{p,p} \leq \dots \leq d_{p,p}$). The estimate of the variable value for the pixel p is then expressed as a function of the closest units, each such unit value weighted according to a distance function in a particular feature space. A commonly used function for weighting distances is:

$$w_{(p_i)p} = \frac{1}{d_{(p_i)p}^t} / \sum_{j=1}^k \frac{1}{d_{(p_j)p}^t} \quad (1)$$

with $t=2$. The estimate of the variable m for pixel p is then:

$$m_p = \sum_{i=1}^k w_{(p_i)p} m_{(p_i)}, \quad (2)$$

where $m_{(p_i)}$, $i=1, \dots, k$, is the value of the variable m in sample plot i corresponding to the pixel $p_{(i)}$, which is the i th closest pixel (of “known” pixels) in the spectral space to the pixel p (Tomppo 1997a).

This estimation procedure is used on an operational basis in the Finnish national forest inventory. Even though the procedures involved have been documented, the analysis of the behavior and quality of the estimation has not been explored in depth. Tomppo (1997b) reported that a method for error evaluation for these methods was under development.

Nilsson (1997) conducted a simulation study to evaluate the kNN for forest volume estimation. His results showed a requirement of at least one plot per 26 km² (0.00038 plots/ha) in the area covered by the simulated forest map. He recommended using 5–10 spectrally nearby samples (neighbors) since the marginal decrease in mean square error (MSE) was found to be small when ten or more were used.

Muinonen and Tokola (1990) applied the “reference sample method” (another name for the kNN method) in two phases: a land use classification (forest/nonforest) and the estimation of forest parameters (growing stock). The estimation procedure was developed for an area of 4000 ha using 1318 plots (0.3295 plots/ha). They concluded that their estimates of volume were unsatisfactory due to several factors, including a sample distribution of plots lacking a good representation of the variation of the forests in the inventory area, and a low correlation between forest variables and satellite data.

Tokola et al. (1996) examined the reference sample method to compare different satellite data, distance metrics, and weighting functions using an average of 0.00214 plots/ha. Considering 10–15 neighbors to be a suitable number of nearest plots for plot variable estimation, they found relative root MSEs larger than 60% of the mean for their best estimate of volume. Improvements would be expected as more pixels with known field values were included.

Moeur, Crookston and Stage (1995) and Moeur and Stage (1995) described the “most similar neighbor” (MSN) modeling procedure. For a plot with unknown field measures, the procedure chooses the most similar plot from among a set of plots detailed to act as stand-in, from which to impute the missing information. The stand-in plot is chosen on the basis of similarity measures that summarize the multivariate relationships between low-resolution indicators and detailed second-phase sample attributes. The similarity function is developed using canonical correlation methods. This inference procedure falls into the framework of kNN procedures with $k=1$. Under these conditions, the MSN estimates retain the full range of variation of the observed data, as well as preserve the natural variability of the data. A possible disadvantage of using MSN is that the predictions are limited to the range of observations in the original sample, therefore, it is important to have a representative sample of the population of interest (Moeur 1988).

2.2. Image classification for mapping

Holmgren and Thuresson (1998) reviewed more than 30 classification studies and summarized their findings as follows: “For a dozen or so forest classes, ranging from regeneration areas to mature stands, classification accuracy is consistently in the range of 65–85% correctly classified pixels.” They also found that including other land uses beside forestry boosts the accuracy of a classification above 90%. Their conclusions are independent of local climatic conditions or remote sensor used.

Among the different multispectral classification approaches described in the literature, the method applied in this research can be described as a hard classification using supervised training. In a hard classification procedure, every pixel is assigned to a single class regardless of its degree of membership (Jensen, 1996). Foody (1999) breaks down the supervised classification process into three steps. First, the class descriptors are generated in the training stage. Second, the descriptors information is used by the classification algorithm to assign each pixel to the class with which it has the greatest similarity. The third step assesses the accuracy of the classification.

Various assignment rules have been utilized in supervised classification. The most frequently used is the maximum likelihood classifier. In this parametric approach, the identity of the field cover types is known a priori, and every pixel is evaluated and assigned to the class for which the

likelihood of being a member is maximum. In order to obtain optimum classification using this probabilistic procedure, it is necessary to assume multivariate normality of the variables involved in the classification.

Horler and Ahern (1986) analyzed the capability of classification systems based on Landsat Thematic Mapper (TM) data to separate forest classes. Their study area was located in Canada in a region similar to the Great Lakes. For forest cover type discrimination, they determined that TM bands 3, 4, and 5 contained most of the information.

There are a few examples of the use of Landsat imagery for classification purposes in the northern Lake States (Minnesota, Wisconsin, and Michigan). Hopkins, Maclean and Lillesand (1988) conducted a preliminary assessment of the Landsat TM data for classifying forests under Lake States conditions (northern Wisconsin). Using a supervised maximum likelihood approach, for nine forest cover type classes and 13 nonforest classes, they reported a 85% overall accuracy (OA) and a 69% accuracy for the nine forest classes. They concluded that Landsat TM imagery also has potential for further detailed classification.

Moore and Bauer (1990) analyzed how forest and sensor characteristics affect the classification accuracy of Minnesota forest cover types. They compared Landsat MSS and TM data on a pixel-by-pixel basis using the maximum likelihood classifier. Using wall-to-wall (pixel-by-pixel) and “test field” accuracy they concluded that TM data performed better than MSS data for forest cover type classification, with a 15–20% increase in accuracy. They concluded that the highly mixed vegetation of north central Minnesota necessitates the collapse of many potential cover types into a few broad classes.

Bauer et al. (1994) developed and implemented a two-stage sampling approach for classification and estimation of forest cover types in northeastern Minnesota. Six forest and five nonforest classes were classified integrating sampling, image processing, and estimation procedures. They used a combination of supervised and unsupervised training referred to as “guided” clustering for classification. Using this approach, they consistently obtained superior results over any other supervised or unsupervised classification method tested. Overall classification accuracy ranged from 64% to 80% for the 11 target classes.

Wolter, Mladenoff, Host, and Crow (1995) provided an example of the use of a multistate approach for forest cover types classification in the northern Lake States region. They first acquired satellite imagery to match specific phenological stages for key forest species. Subsequently, they classified 13 forest cover type classes in a nine-stage knowledge-based and maximum likelihood combined approach. The overall classification accuracy reported was 83%, and the forest classification accuracy was 80%.

The kNN method is a nonparametric classifier in which there are no assumptions about the distributions of the variables involved in the classification (Hardin, 1994).

Unlike their parametric counterparts, they do not summarize the training classes prior to the pixel assignment step. Instead, the information for all training pixels is stored, and the unlabeled pixel is classified “taking a vote” among the neighboring training pixels. These methods have not been widely used, perhaps because they are computationally intensive for practical problems. Hardin (1994) summarized a substantial body of literature regarding the statistical characteristics of nearest-neighbor rules and stated “when the proportion of pixels in each training class is identical to the actual proportion of each class in the population, the k-NN rule is a maximum-likelihood classifier.”

Hardin (1994) also compared the performance of parametric and nonparametric classifiers, particularly nearest neighbor rules. Ten different pixel assignment rules, six of them nonparametric, were applied to five test images of 15,000 pixels each. This study concluded that the neighborhood based classifiers, in particular, the distance weighted neighbor classifier, are superior to the best parametric classifiers when the training sets are large and contain the same class proportions as the population to be classified. When this condition is severely violated, there was not a clear advantage in using nearest neighbor pixel assignment rules.

Most of the kNN examples available in the forestry literature deal with the estimation of basal area and volume. In many of these applications, the procedures involved have been documented, but analysis of the behavior and quality of the estimation has not been explored in depth. Less effort has been devoted to kNN for forest cover type mapping.

3. Methods

For the applications of kNN in this study, it was deemed important to assess a number of factors affecting the behavior and quality of estimation. The following sections describe these considerations.

3.1. Data preparation

3.1.1. Study area

Estimates of forest basal area, volume, and cover type were developed and examined for St. Louis County, MN. This county is located in the northeastern part of the state and comprises about 1.6 million ha, of which approximately 82% is forest (Miles, Chen, & Leatherberry, 1995). Many different forest types are included in this area dominated by Aspen–Birch and Spruce–Fir associations. Table 1 shows the distribution of cover types as determined from the FIA reference data.

3.1.2. Field data

The study area is part of the FIA Aspen–Birch Unit. During Minnesota’s fifth forest inventory (1986–1991),

Table 1

Cover types distribution and codes for St. Louis County as determined from FIA reference data

Code	Cover type class	Percentage of forest area
JP	Jack Pine	3.4
RP	Red Pine	2.0
WP	White Pine	0.8
BF	Balsam Fir	8.7
BS	Black Spruce	17.1
NWC	Northern White Cedar	2.9
T	Tamarack	4.5
WS	White Spruce	1.2
EAS	Elm–Ash–Soft Maple Group	5.6
MBB	Maple–Birch–Beech Group	3.5
A	Aspen	41.8
PB	Paper Birch	6.8
BP	Balsam Poplar	1.7

2582 forested plots, each consisting of a 10-point cluster, and sampling approximately 1 acre (0.4047 ha), were measured in the county (Anonymous, 1990, 1997a). Using the variables land use and change in land use, the FIA database was queried to obtain plot cover type classifications for forest land uses in the study area (excluding reserved forest, e.g., designated park and wilderness areas that did not contain plots). Conditioning on the study time frame and available imagery, 1195 plots were identified. The resulting sampling intensity was approximately 0.0009 plot/ha. Fig. 1 shows the distribution of these plots. Additionally, using the variables plot location, ground land use, and disturbance (from image analysis), the FIA database was queried to obtain basal area (trees ≥ 2.5 cm DBH) and net cubic wood volume (trees ≥ 12.7 cm DBH) per hectare for 965 undisturbed forest plots. These were plots measured and judged “undisturbed” in the timeframe covered by the three dates of imagery used (see below). The resulting sampling intensity was approximately 0.0006 plots/ha. These forest plot samples are a subsample of those shown in Fig. 1 and were used for estimation of basal area and volume.

Hansen and Hahn (1992) described the procedures used by North Central Forest Experiment Station’s FIA in determining forest type. The algorithm used to classify plots is a function of the species, stocking, and size of the trees, and the physiographic class of the site. A general description of the procedure is as follows. Stock values are summed for all live trees into type groups based on species. After all trees are combined using physiographic class, the forest type is determined by comparing total stocking in the combined type groups. In the instances in which several type groups are compared, the predominant group is selected by plurality. In particular, for the Lake States, the first step in the algorithm compares total stocking in hardwoods with total stocking in softwoods. If at least one-half of the total stocking is in softwoods, one of the softwood types is assigned, or vice versa.

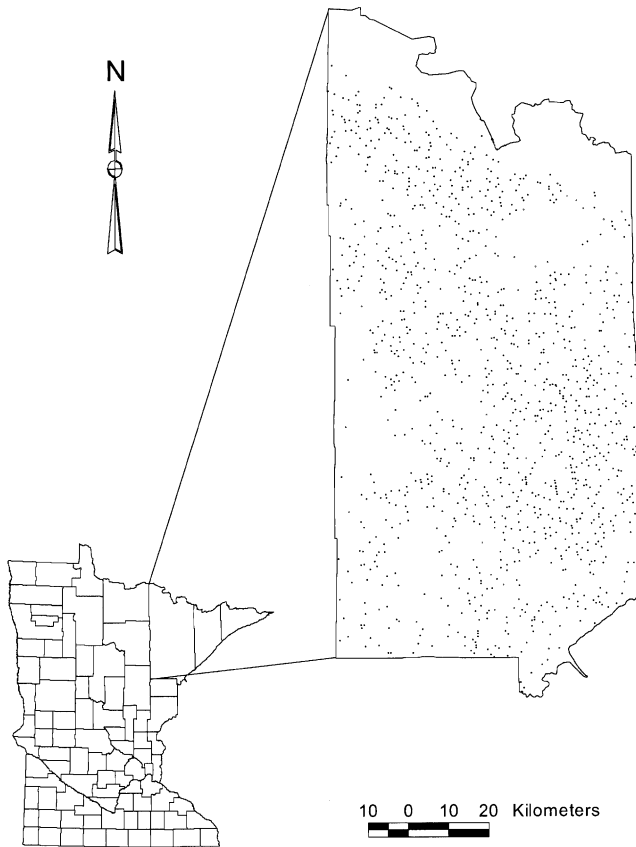


Fig. 1. Distribution of FIA plots ($n=1195$) in St. Louis County, MN.

Mixed hardwood–softwood types are not recognized in this procedure.

The plot locations and the values of basal and volume from the FIA sampling plots were combined with information of the multitemporal remote sensing analysis and ancillary data described below. In Minnesota's fifth forest inventory, plot locations were determined using aerial photos and USGS 1:24,000 quad maps (Hackett 1986).

3.1.3. Landsat TM data

St. Louis County is covered by portions of two Landsat TM images (rows 26 and 27 in path 27). To ensure compatibility between images and with the ground data, each image was rectified and georeferenced to the UTM system using the following parameters: Spheroid GRS 1980, datum NAD83, and zone 15. The resampling method was nearest neighbor with a 30×30 m pixel size. A mosaic was constructed by subsetting the St. Louis County portion of the rectified images.

The above procedure was repeated for each of the three dates included in the study: an early fall scene from September 25, 1987; a winter scene from March 3, 1988; and a summer scene from June 7, 1988. These dates were chosen to include diverse phenological stages of the vegetation in the region and to avoid the presence of

clouds. A single multitemporal image (consisting of 18 bands) was constructed using TM bands 1, 2, 3, 4, 5, and 7 from each date. Band 6 was not used because of its coarse resolution.

3.1.4. Ancillary data

Information provided by an ecological classification system (ECS) for the state (Anonymous, 1998b) was used as an ancillary source of information. This classification system attempts to integrate climatic, geologic, topographic, hydrologic, soil, and vegetation data in a map. Nine ECS subsection level map classes are included in St. Louis County. These are broadly defined and mapping units vary in size from tens to thousands of square miles (Albert, 1995).

The ancillary information was included in our study using a penalty value function, i.e., if two neighbors are located at a similar distance from the pixel to be classified, but one does not belong to the same ECS class, its distance is then penalized. This approach gives preference to the plots belonging to the same ancillary data class.

Using the FIA plot location as a common relational feature between the satellite image and the ground information, we combined all the above noted information in a common file. The contents of the resulting file were x , y coordinates for plot location, 18 Landsat TM bands, ECS class, basal area (m^2/ha), volume (m^3/ha), and cover type for each FIA plot.

3.2. Analysis

3.2.1. Distance metric

Distances between neighbors were computed using two different distance metrics, Mahalanobis and Euclidean. The general expression for the distance between pixel p (to be classified) and pixel p_i , for which the ground data is known, was given earlier as Eq. (3):

$$d_{p(p_i)} = \sqrt{\sum_{j=1}^{nf} (x_{p,j} - x_{(p_i),j})^2} \quad (3)$$

where $x_{p,j}$ = digital number for the feature j , nf = number of features in the spectral space. In the above-mentioned consideration of ancillary data, we also added an arbitrary penalty to the equation, i.e., a value applied to units not belonging to the same ancillary information class as pixel p .

Applying Eq. (3) to our data is equivalent to computing Euclidean distance. To calculate the Mahalanobis distance among neighbors, we applied this same equation in a transformed space. Eq. (4) describes the transformation, based on the square root matrix, used to map Mahalanobis space to Euclidean space:

$$\mathbf{y}^t = \mathbf{x}^t \mathbf{P} \mathbf{\Lambda}^{-1/2} \quad (4)$$

where $\mathbf{y}$ = transformed vector, $\mathbf{x}$ = original vector, $\mathbf{P}$ = matrix of column eigenvectors of $\mathbf{x}$'s variance–covariance matrix

(S) and, $\Lambda^{-1/2}$ = diagonal matrix of the inverse square root eigenvalues of S.

3.2.2. Neighbor's weighting function

In order to investigate the relative importance of the neighbors in constructing estimators, the weight of the pixel p_i in estimating a variable on pixel p was computed using three different weighting functions: (a) equal; (b) inversely proportional to the distance; and (c) inversely proportional to the square of the distance. These weights are obtained from Eq. (1) by choosing $t=0, 1$, or 2 , respectively.

For basal area and volume estimation, once the distances among neighbors and their weights in the estimation were calculated, the kNN method estimator was applied to each pixel. The estimator of the variable m for the pixel p is then obtained from Eq. (2). For cover type estimation, the kNN weighted mode estimator was applied to each pixel.

3.2.3. Image and feature space enhancement

Two different image enhancement techniques were applied — spatial filtering and computation of vegetation indices (VIs). The goal of using spatial filters was to encompass and approximate the 10-point cluster plot layout used in FIA inventory. For this purpose, 4×3 pixel filters were constructed, using (1) mean and (2) median band values. The results using filtered information were then compared with the results obtained using the TM data for the plot center pixel without any filtering. We note that exact locations are problematic. There is location inaccuracy in FIA plots and their georeferencing. In addition, there is inaccuracy in the satellite data from registration and rasterizing the sensor detail to pixels, plus differences in registration between dates of imagery. In this study, anecdotal evidence suggests point 1 of the 10-point FIA

cluster was typically georeferenced to within 30 m of its true position.

Additionally, new features were generated using the tasseled cap transformation and other VIs. For the tasseled cap transformation, brightness, greenness, and a “third” component were computed for each individual image using the coefficients implemented in the ERDAS Imagine software package (Anonymous, 1997b). The set of “unique” VIs recommended by Coppin and Bauer (1994) and tested in this project were: TM4/TM3, TM4/(TM4 + TM3), TM2/TM4, TM5/TM4, and TM7/TM4. The performance of these new features was compared to the performance of the 18 original features or spectral bands.

3.2.4. Feature weighting parameters

Not all the features in the feature space share the same influence in the prediction of a forest variable for a given pixel. Assuming that there exists a linear combination of features that can provide the best result, additional weights were computed and applied to the original features. This weighting parameter development was developed by applying the downhill simplex optimization method developed by Nelder and Mead (1965), adapting the *amoeba* “recipe” from Press, Teukolsky, Vetterling, and Flannery (1994) to this crossvalidation problem. The resulting expanded form of Eq. (3) is:

$$d_{p(p_i)} = \sqrt{\sum_{j=1}^{nf} a_j^2 (x_{p,j} - x_{(p_i),j})^2} \quad (5)$$

where a_j = weighting parameter for the feature j . We note that this optimization brings the kNN approach closer to linear regression approaches in terms of the effective use of features or predictor variables.

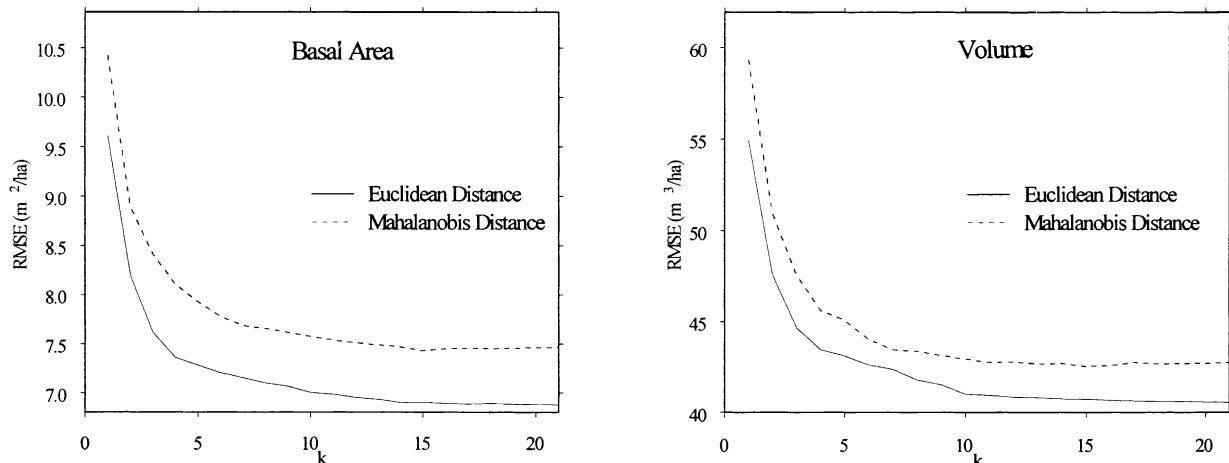


Fig. 2. Basal area and volume RMSE for two different distance metrics and different numbers of neighbors (k).

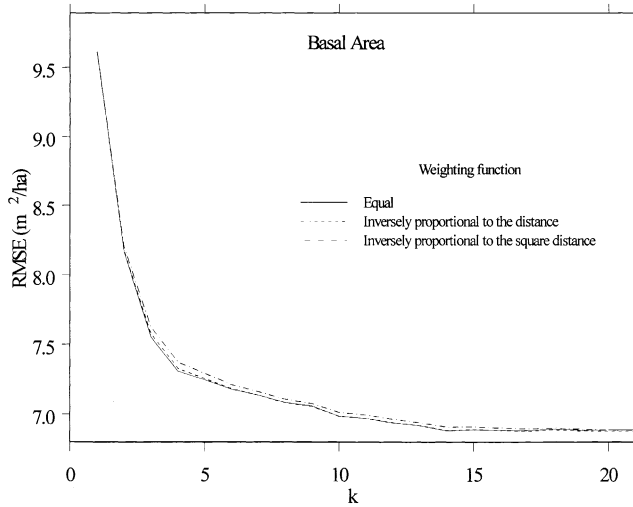


Fig. 3. Basal area RMSE for three different neighbor weighting functions and different numbers of neighbors (k).

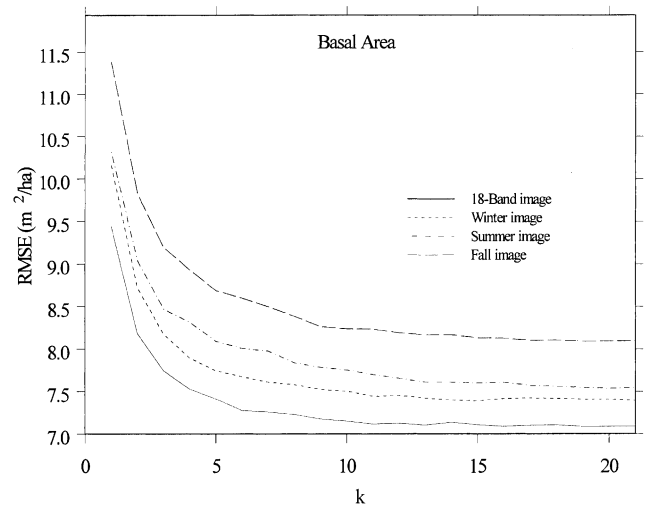


Fig. 5. Basal area RMSE for the 18-band image compared with the single date images for different numbers of neighbors.

3.2.5. Prediction error estimation — basal area and volume estimation

After obtaining an independent estimate based on the kNN for each one of the pixels in the training set, we proceeded to evaluate the results. The estimation was evaluated using prediction error, which measures how well a model predicts the response value of a future observation. Burk (1990) compared different resampling techniques to estimate true prediction error. He suggested that at a minimum a resampling procedure should be considered for model evaluation, among these techniques: “data splitting,” crossvalidation, jackknifing, and bootstrapping. Consequently, we estimated prediction error in several ways.

3.2.5.1. Direct calculation. For every trial, the accuracy of our estimates of basal area and volume were examined using the root mean square error (RMSE) (Eq. (6)):

$$\text{RMSE} = \sqrt{\sum_{i=1}^n (y_i - \hat{y}_i)^2 / n} \quad (6)$$

where y_i is the variable of the interest on the i th observation and $\hat{y}_i$ is the predicted value (in this case from applying the kNN prediction rule). However, estimating the *true prediction error* of a model using the same data used to fit it tends to be too “optimistic,” since the model is fine-tuned to that data. In other words, the test sample is the same

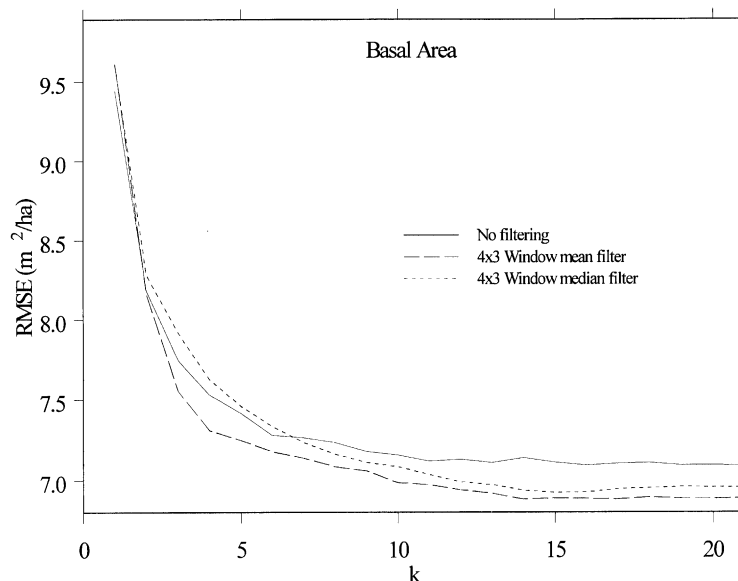


Fig. 4. Basal area RMSE for three different filters and different numbers of neighbors (k).

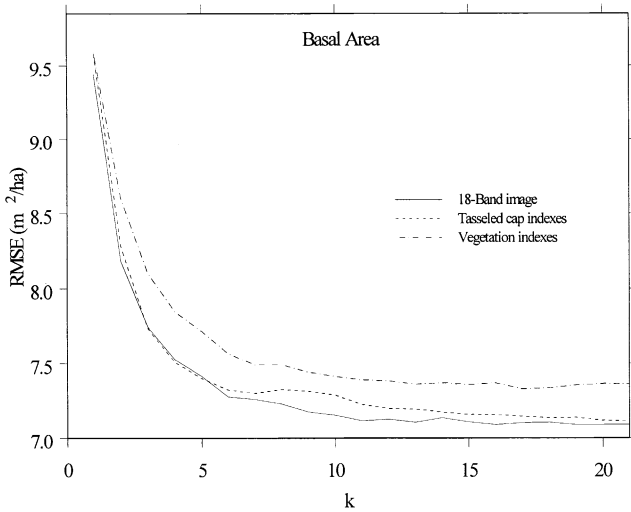


Fig. 6. Basal area RMSE for the 18-band image compared with image composites of vegetation indices for different numbers of neighbors (k).

as the training sample and the estimator tends to be downwardly biased. Estimates of prediction error obtained in this way are aptly called apparent error estimates (Efron & Tibshirani, 1993).

3.2.5.2. Crossvalidation. Gong (1986) describes cross-validation as omitting training sample units one by one. As such, it is an evaluation technique that mimics the use

of independent data. For each omission, apply the prediction rule (kNN in this case) to the remaining sample and summarize the error that the rule makes when it predicts the omitted unit. In totality, we apply the prediction rule n times and predict the outcome for n units. The mean of errors made in these n predictions is the crossvalidation estimate of error (Eq. (7)):

$$\text{RMSE}_{\text{CV}} = \sqrt{\sum_{i=1}^n (y_i - \hat{y}_i^{-i})^2 / n} \quad (7)$$

where $\hat{y}_i^{-i}$ is the predicted value of the i th observation using a prediction rule fitted without considering observation i . This estimate of prediction error is nearly unbiased, but can be highly variable (Efron & Tibshirani, 1997).

3.2.5.3. Bootstrapping. The simplest bootstrap method generates B bootstrap samples, estimates the prediction rule on each, and then applies each rule to the original sample, therefore providing B estimates of prediction error. A more refined version of the bootstrap approach is “leave-one-out” bootstrap ($\text{RMSE}_{\text{LOOB}}$). This is a smoothed version of RMSE_{CV} in which the error at point i is only estimated from bootstrap samples that do not contain the point i .

Efron and Tibshirani (1997) also designed the 0.632 bootstrap estimator (Eq. (8)), to correct the upward bias in $\text{RMSE}_{\text{LOOB}}$ by averaging it with the downwardly biased estimator apparent RMSE. The coefficients $0.632 \approx 1 - (1 - 1/n)^n$ and 0.368 were suggested by an argument

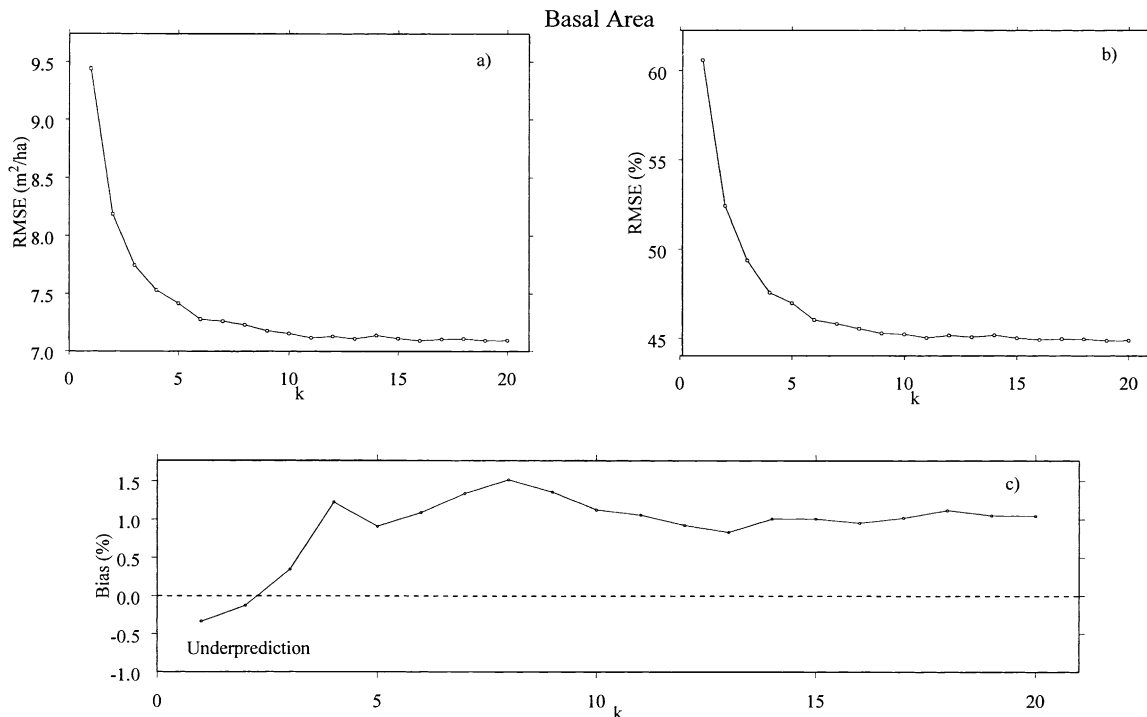


Fig. 7. Basal area estimation with kNN: (a) basal area RMSE with different number of neighbors; (b) basal area relative RMSE; and (c) bias when applied for different number of neighbors.

based on the fact that bootstrap samples are supported on approximately $0.632n$ of the original data points:

$$\text{RMSE}_{0.632} = 0.368\text{RMSE} + 0.632\text{RMSE}_{\text{LOOB}} \quad (8)$$

3.2.5.4. Confusion matrices. Besides the use of RMSE as a global estimator of error, we also built confusion matrices for basal area and volume. The basal area confusion matrix consisted of four arbitrarily specified $10 \text{ m}^2/\text{ha}$ basal area classes and the volume confusion matrix consisted of five $40 \text{ m}^3/\text{ha}$ classes. Producer's and user's accuracy as described by Stehman (1997) was computed for each one of these confusion matrices and used as supplemental information in evaluations for each trial.

3.2.6. Prediction error estimation — cover type estimation

For cover type, the precision of the classification was examined using the OA of the confusion matrix (Congalton, 1991; Stehman, 1997) or the error rate (Err). Err indicates the disagreement between a predicted value $\hat{y}$ and the actual response y in a dichotomous situation such as, “ y does or does not belong to class i ,” with values 0 or 1 (Efron, 1983). Because of the complementary nature of the OA, we can define (Eq. (9)):

$$\text{OA} = 1 - \text{Err} \quad (9)$$

where

$$\text{Err} = \sum_{i=1}^n (y_i - \hat{y}_i) / n$$

This is a special case of the MSE for an indicator variable.

These estimators were preferred over the usual Kappa estimator because of the assumptions behind the latter in the

definition of chance agreement. In addition, Kappa penalizes confusion matrices in which row and column marginal proportions are similar (Stehman, 1997).

The crossvalidation estimate of the MSE for an indicator variable is (Eq. (10)):

$$\text{Err}_{\text{CV}} = \sum_{i=1}^n (y_i - \hat{y}_i^{-i}) / n \quad (10)$$

where $\hat{y}^{-i}$ is the predicted value of the i th observation using a prediction rule fitted without considering observation i . Additionally, we employed the $0.632+$ bootstrap estimator ($\text{Err}_{0.632+}$), which Efron and Tibshirani (1997) found outperformed crossvalidation in a catalog of 24 experiments. The characteristics of this estimator are (Eq. (11)):

$$\begin{aligned} \text{Err}_{0.632+} = & 0.368\text{Err} + 0.632\text{Err}_{\text{LOOB}} \\ & + (\text{Err}_{\text{LOOB}} - \text{Err}) \frac{(0.368)(0.632)\hat{R}}{1 - 0.368\hat{R}} \end{aligned} \quad (11)$$

where $\hat{R}$ is the relative overfitting rate of a classification rule. In a highly overfit rule, Err tends to be small.

Programming code for the procedures described above was developed in C and is available from the authors at the University of Minnesota upon request.

4. Results and discussion

4.1. Basal area and volume estimation

Analyses were first conducted for the dependent variables basal area and volume per hectare. In most cases, the prediction rule with the lowest crossvalidation RMSE was

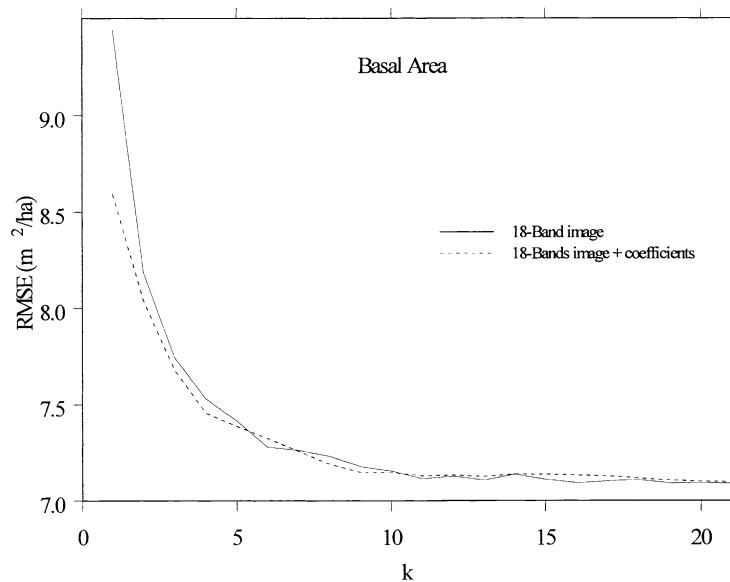


Fig. 8. Basal area RMSE for the 18-band image compared with the same image using coefficients generated using the downhill simplex optimization method for different numbers of neighbors (k).

regarded as the best. Given the nature of the kNN technique, the estimates for both variables were constructed using the same neighbors. Given this procedure plus the high correlation between these variables, the results for volume were very similar to those for basal area. Consequently, unless specified, the results for volume are essentially the same as those for obtained for basal area.

4.1.1. Distance metric evaluation

Fig. 2 shows the RMSE values obtained in the estimation of basal area and volume when Mahalanobis and Euclidean distance are used. This illustrates the typical behavior of the prediction errors associated with the kNN estimator. There is a rapid early increase in the precision of the technique with the addition of the first few neighbors, thereafter the marginal gains diminish and precision levels off. Even though there is a well-known correlation among TM band values, the use of Mahalanobis distance did not benefit the quality of the estimation in these trials. We note that this is contrary to the results reported by Nilsson

(1997). The Euclidean distance metric produced RMSE results at least 5% smaller than those for the Mahalanobis metric for any number of neighbors. Given this result, we set the Euclidean distance as the distance metric for use in subsequent trials.

4.1.2. Neighbor's weighting functions

The results of applying the three different weighting functions for basal area estimation are shown in Fig. 3. There was no evidence in support of the functions in which significantly more weight was assigned to the closest neighbor. The simple mean estimator worked best. In order to take advantage of weighting functions such as inversely proportional to the distance or inversely proportional to the square of distance, it is necessary to have an identifiable close neighbor and some distance between this and the next one. This was not the case. It was apparent that instead of having one spectrally close neighbor, in most cases, there were several close neighbors. However, this could be due in part to the FIA plot grid layout, which spaced plots regularly

Table 2

Estimation summary for kNN classification of basal area when $k=1$ and $k=9$

(a) General information						
Mean basal area:	15.64	Standard deviation of basal area:	8.88	Range:	0.04–51.55	
Distance function:		Euclidean				
Weighting function:		Equal				
Number of plots:	965					
Number of bands:	18					
Filter:	None					
Penalty value:	None					
Coefficients:	None					
(b) Basal area estimation ($k=1$)						
<i>RMSE estimators (m^2/ha)</i>						
Crossvalidation	9.44372	Reference data (basal area class, m^2/ha)				
Confusion matrix		0–10	>10–20	>20–30	>30	User's accuracy
Classified	0–10	156	91	27	5	0.55914
	>10–20	104	170	99	17	0.43590
	>20–30	22	98	91	23	0.38889
	>30	7	12	28	15	0.24194
Producer's accuracy		0.539792	0.458221	0.371429	0.25	
(c) Basal area estimation ($k=9$)						
<i>RMSE estimators (m^2/ha)</i>						
Crossvalidation	7.17762	Reference data (basal area class, m^2/ha)				
Confusion matrix		0–10	>10–20	>20–30	>30	User's accuracy
Classified	0–10	121	49	9	3	0.66484
	>10–20	155	247	98	18	0.47683
	>20–30	13	75	135	38	0.51724
	>30	0	0	3	1	0.25000
Producer's accuracy		0.418685	0.665768	0.55102	0.016667	

across the landscape. By contrast, in the Finnish national forest inventory, plots are grouped closely within clusters, with a substantial distance between clusters. Thus, for subsequent trials, Eq. (1), with $t=0$, was used to allocate equal weights to neighbors.

4.1.3. Image enhancement — filtering

The results of the image filtering trials are shown in Fig. 4. There is a slightly more precise estimation using the 4×3 pixels window mean filtering when more than two neighbors are used in the estimation. The maximum gain in precision due to the use of a filter was approximately 3%. However, if we compare this gain with the possible loss of information due to the smoothing effect of filtering, the effort is questionable. Since the filter's smoothing effect could mask small differences between pixels and since the kNN is based on detecting the spectral differences among them, filtering does not seem very attractive.

The argument for using a multiple pixel window to address plot location error is also debatable. Part of the motivation behind using a filter window was to fit the FIA plot layout used in Minnesota's fifth forest inventory. However, given a tradeoff between smoothing and location error, the former seems easiest to avoid. Consequently, the basal area and volume estimation trials were completed without a filter.

4.1.4. Feature space enhancement — multitemporal vs. single-date approach

Working with an 18-band image can be computationally very intensive. In an effort to reduce the dimension of the estimation problem, we compared the performance of the 18-band image against using the single-date images independently. The results of these trials are shown in Fig. 5. None of the single dates outperformed the use of the multitemporal approach. Using all three dates (18 bands) produced 5% less RMSE on average than the closest competitor. The winter date by itself produced the second-best estimation. The explanation for this result is that variables like basal area and volume appear to be more associated with the percent occupancy of a pixel rather than with the specific vegetation radiance values, contrary to a classification problem.

4.1.5. Feature space enhancement — tasseled cap and other VIs

The next step was to compare the performance of the 18-band image against image composites of the tasseled cap indices and other VIs. Even though these indices often provide “new” information (specifically ratios), none provided better results than using the 18-band image (Fig. 6). The tasseled cap index, however, has a place as a dimension reduction technique. Since it is based on principal components analysis, it does not create information not present in the original image, but it may help to simplify or explain the spectral–radiometric–temporal feature space.

4.1.6. Feature space enhancement — ancillary information

The use of ancillary ECS information (as a penalty value in the distance function) also did not improve results. For any value of penalty tested the results in RMSE were the same. This is probably due to the coarse resolution of the ancillary information available. Additionally, the correlation between the ancillary information classes and basal area or volume is too small for them to be helpful for stratification. Finally, The FIA plot sampling intensity was likely too low for nearest neighbors to consistently be found within a short distance, i.e., within the same ECS class.

4.1.7. Number of neighbors

The errors in kNN estimations for basal area using Euclidean distance, equal weighting among neighbors, unfiltered data and 18 bands are depicted in Fig. 7. Parts (a) and (b) in this figure show the behavior of RMSE and relative RMSE. It is clear that there was a rapid early gain in overall precision with the addition of neighbors. The values for the RMSE dropped approximately 14% when the number of neighbors was increased from one to five. After the number of neighbors reached nine, the marginal increase in precision was smaller than 0.5%. This result agrees with the findings of several other authors who reported this stability point between 10 and 15 neighbors (Nilsson 1997; Tokola et al., 1996)

Fig. 7c shows the nearly unbiased behavior of the kNN estimator. For all values of k neighbors, the bias was smaller than 1.5%. In particular, when $k=1$, the bias was smaller than 0.4%. Even though the magnitude of this bias is small, it requires some explanation. Consequently, we examined the behavior of the basal area RMSE (Fig. 8) with respect to the basal area confusion matrix producer's accuracy in Table 2. In this case, the classification attempted the grouping of

Table 3
Coefficients for the TM bands of the 18-band image obtained using optimization for basal area and cover type estimation

Image date	TM band	Coefficient (a) for basal area	Coefficient (a) for cover type
25 Sep 87	1	1.214254159	0.658637292
	2	1.280423902	1.941597448
	3	2.190238828	−3.947417041
	4	1.165821366	1.167866404
	5	−0.133686351	1.858930644
	7	1.256088889	2.239643403
	7	1.256088889	2.239643403
3 Mar 88	1	2.053957268	−0.234960101
	2	0.983996425	0.697096680
	3	0.628646185	1.886745301
	4	1.270501984	3.136860104
	5	1.352639611	1.485621708
	7	1.191953333	1.708555389
	7	1.191953333	1.708555389
7 Jun 88	1	0.951554179	2.020878952
	2	4.766569659	2.464458578
	3	1.094327795	2.228461559
	4	0.920436953	0.345704069
	5	0.476908882	0.241766177
	7	0.712703836	2.876920345
	7	0.712703836	2.876920345

basal area into four classes. Although increasing k significantly reduces the overall RMSE, this step also leads to a large reduction in the producer's classification accuracy for the two extreme classes. Thus, the bias increases as k increases for the individuals in the two extreme classes basal area classes (0–10 and >30 m²/ha); these are pulled towards the mean and most populated classes.

Considering only the largest class (>30 m²/ha), increasing the number of neighbors from one to two causes a reduction of 60% (from 25% to 10% producer's accuracy) in the accuracy for classifying the plots with the largest basal area. When we include more than seven neighbors the producer's accuracy reaches a minimum. These results are illustrated in Table 2 under producer's and user's accuracy

Table 4
Estimation summary for kNN basal area and volume estimation when $k=1$

(a) General information							
Mean basal area: 15.64		Standard deviation of basal area: 8.88		Range: 0.04–51.55			
Mean volume: 61.62		Standard deviation of volume: 50.50		Range: 0.00–277.41			
Distance function:		Euclidean					
Weighting function:		Equal					
Number of plots:		965					
Number of bands:		18					
Filter:		None					
Number of neighbors:		1					
Penalty value:		0					
Coefficients:		Yes					
Number of Bootstrap samples:		200					
(b) Basal area estimation							
<i>RMSE estimators (m²/ha)</i>							
Apparent	0.0						
Crossvalidation	8.59548						
Leave-one-out bootstrap	9.09887						
Bootstrap 0.632	5.75049						
		Reference data (basal area class m ² /ha)					
Confusion matrix		0–10	>10–20	>20–30	>30	User's accuracy	
Classified	0–10	160	97	28	3	0.55556	
	>10–20	106	173	94	15	0.44588	
	>20–30	22	94	98	30	0.40164	
	>30	1	7	25	12	0.26667	
Producer's accuracy		0.55363	0.46631	0.4	0.2		
(c) Volume estimation							
<i>RMSE estimators (m³/ha)</i>							
Apparent	0.0						
Crossvalidation	51.6084						
Leave-one-out bootstrap	53.3001						
Bootstrap 0.632	33.6857						
		Reference data (volume class, m ³ /ha)					
Confusion matrix		0–40	>40–80	>80–120	>120–160	>160	User's accuracy
Classified	0–40	259	91	41	12	8	0.63017
	>40–80	92	79	59	21	5	0.30859
	>80–120	38	55	33	36	11	0.19075
	>120–160	13	16	35	13	13	0.14444
	>160	3	8	10	7	7	0.20
Producer's accuracy		0.63951	0.31727	0.18539	0.14607	0.15909	

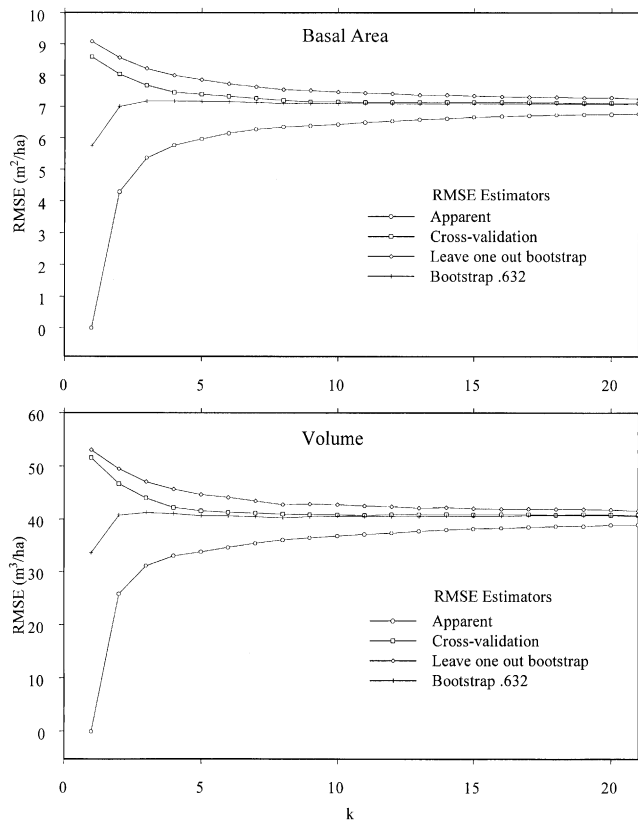


Fig. 9. Prediction error estimation using different statistical procedures for different numbers of neighbors and 50 bootstrap samples.

for basal area when $k=1$ and $k=9$. Note from this table that with a perfect classification all plots would fall on the diagonal. This result implies that increasing the number of neighbors produces an undesirable reduction of variability in predicted values, especially if the goal of using kNN is map production. Clearly, this problem is reduced with use of fewer neighbors and it disappears using only one neighbor.

These results show a trade-off between map accuracy and global objectives for the kNN method. For a global error estimator, the conclusion would be to use $k=9$ neighbors to produce the best estimation. However, the basal area class and map estimation objectives led us to question this conclusion. The decision about how many neighbors to use depends on the objective of the estimation. If the goal is to produce a global estimation for a region then using nine neighbors would be appropriate. However, if the objective is map production, using $k=1$ may be the best option since it retains the full range of variability present in the data.

4.1.8. Weighting parameters

Once it was decided to construct basal area and volume maps using only the nearest neighbor, the next step was to compute coefficients (weighting parameters) for each TM band. To do so, the downhill simplex optimization method was adapted to the crossvalidation problem to minimize RMSE with $k=1$. Weights so determined were then used in computations for estimators based on $k=1, 2, \dots$, etc.

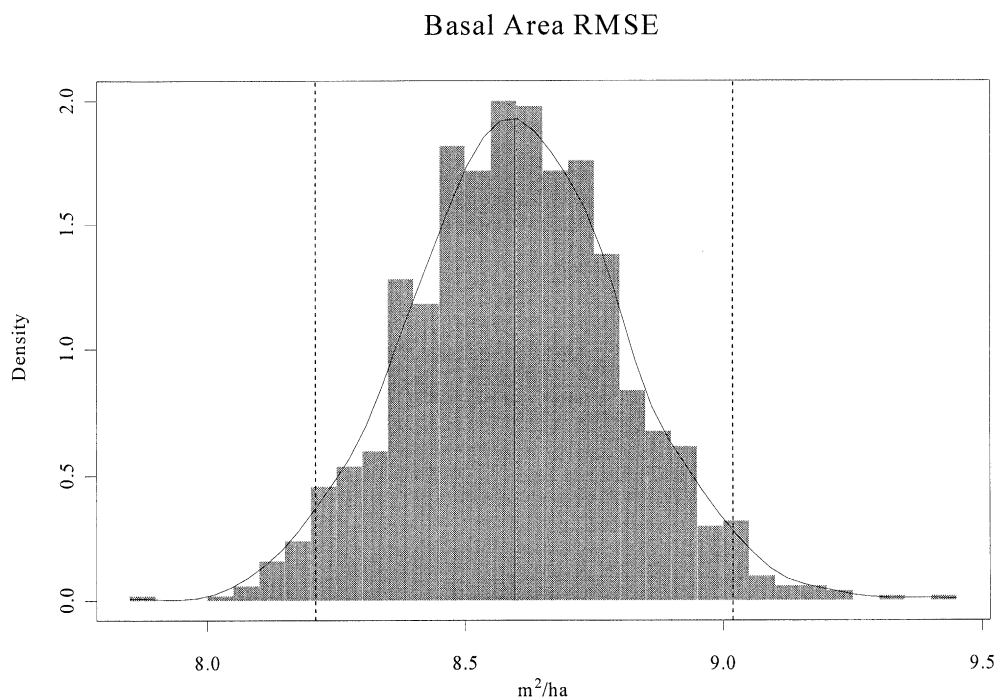


Fig. 10. Empirical 95% confidence intervals for the crossvalidation RMSE of basal area when used for kNN estimation with $k=1$.

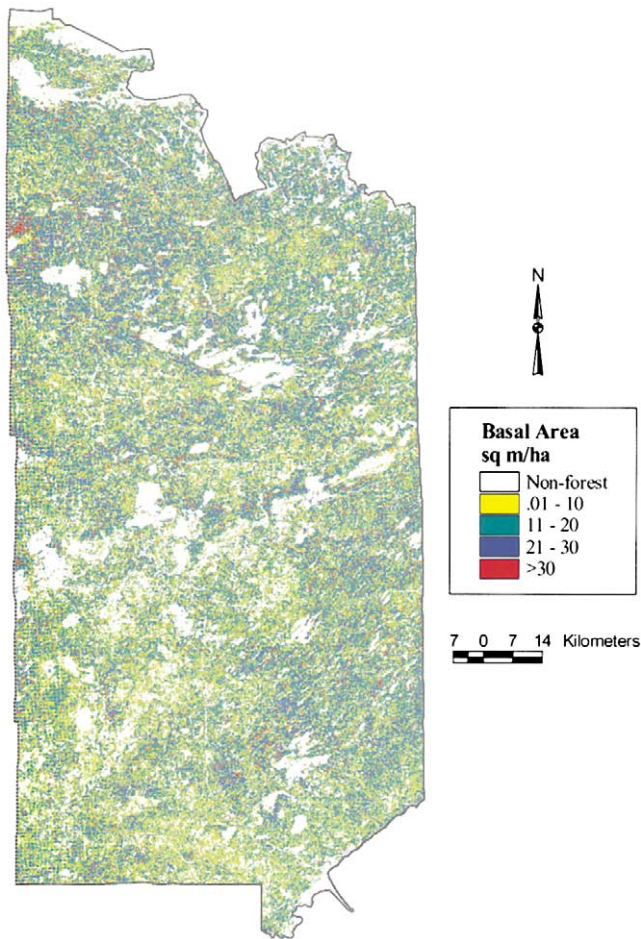


Fig. 11. Basal area map for St. Louis County, MN, using the nearest neighbor estimation ($k=1$).

The coefficients shown in Table 3 were applied in the context described in Eq. (3) to compute the kNN estimates. Table 4 and Fig. 9 compare the performance of the kNN estimator with and without these parameters. Gains with optimization were modest except for $k=1$. We note that this result is supportive of map production. For $k=1$, using the weighting parameters produced a reduction in RMSE of approximately 9%, from 9.4 to 8.6 m^2/ha . Table 4 shows a summary report of the results obtained when applying kNN using $k=1$ for estimating basal area and volume using weighting parameters and 200 bootstrap samples.

4.1.9. Prediction error estimation

The objective of seeking improvements in error estimation through bootstrapping techniques was to obtain an estimate of true error as good as obtained using crossvalidation, but with smaller variance. Fig. 9 shows the behavior of the different prediction error estimation procedures for different numbers of neighbors and 50 bootstrap samples. The 0.632 bootstrap is a nearly unbiased estimator with smaller variability than crossvalidation. It produces good results in almost every case except with overfitted prediction rules. When only the nearest neighbor is used in the estimation the apparent error is zero (obviously an overfitted rule), therefore, $\text{RMSE}_{0.632+}$ becomes $0.632\text{RMSE}_{\text{LOOB}}$ (see Eq. (8)) producing a biased estimation of error. However, the use of bootstrap 0.632 when $k>1$ looks promising.

In order to further characterize results when $k=1$, an empirical distribution of the RMSE was developed for basal area and volume, using simple bootstrapping with 1000 bootstrap samples. The results are shown in Fig. 10. The empirical 95% confidence interval for the basal area

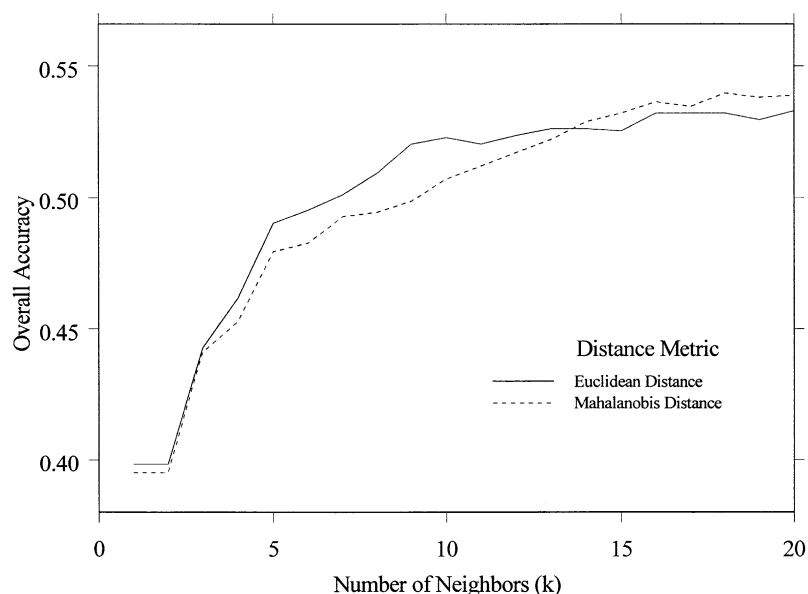


Fig. 12. OA of cover type classification for two different distance metrics and different numbers of neighbors (k).

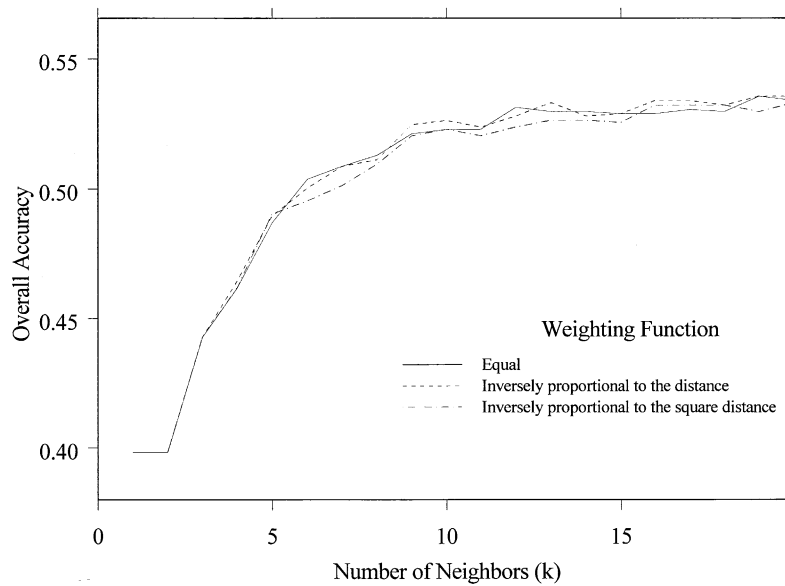


Fig. 13. Overall classification accuracy for three different neighbor weighting functions and different numbers of neighbors (k).

RMSE is (8.209, 9.019) m^2/ha ; for volume, it is (48.68, 54.58) m^3/ha .

4.1.10. Map production

The experience generated in the above trials was then transferred to the construction of thematic maps of forest variables. Using the 18-band composite image, no filtering, Euclidean distance, and the nearest neighbor, basal area (Fig. 11) and volume maps were generated for St. Louis County. The classes considered in these thematic maps

correspond to the same classes examined in the confusion matrices. Note further that the technique produces a map for the northernmost portion of the county (the Boundary Water Canoe Area Wilderness) where there were no actual field plots (e.g., compare the upper portion of Fig. 1 with Fig. 11). While such extensions are likely to be biased, they may still be useful in resource management.

The above results were obtained using imputation at the pixel level with information from 965 undisturbed plots. However, a more challenging scenario for the kNN appli-

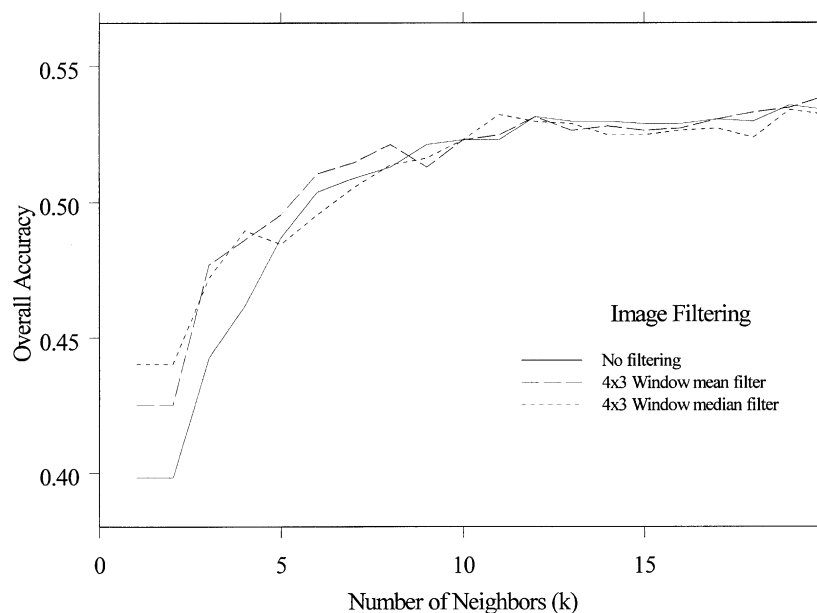


Fig. 14. Overall accuracy of cover type classification for three different filters and different numbers of neighbors (k).

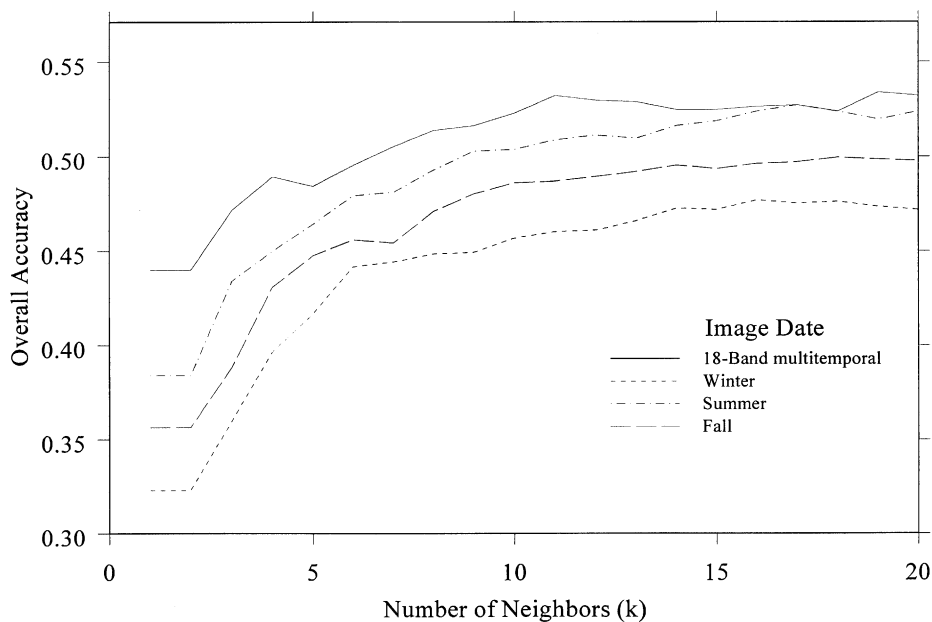


Fig. 15. Overall accuracy of cover type classification for the 18-band image compared with the single date images for different numbers of neighbors (k).

cation would be to consider all disturbance classes in the FIA. In doing so, 329 additional plots were added. These additions were obtained by considering forested plots exhibiting land clearing, harvesting (including thinning), natural regeneration, artificial regeneration, insect or disease damage, and fire. With the inclusion of the disturbed plots, the

number of plots increased considerably in all basal area classes. Subsequently, all estimation procedures were applied to this new sample of 1294 plots. The results for all tests were consistent with the findings described previously. However, a major difference with respect to precision was noted. When $k=1$, the values of the RMSE went

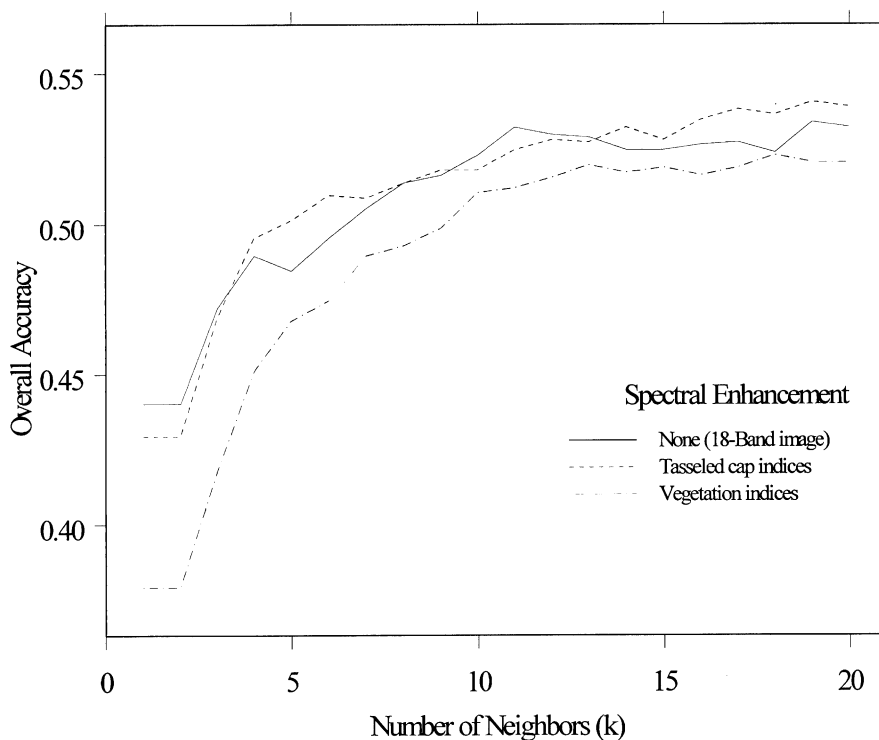


Fig. 16. Overall accuracy of cover type classification for the 18-band image compared with image composites of vegetation indices for different numbers of neighbors (k).

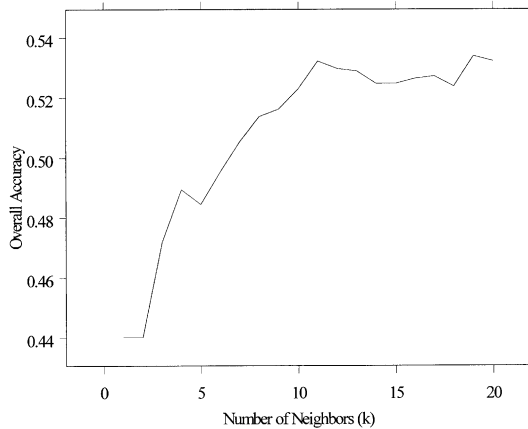


Fig. 17. Overall accuracy of cover type classification with different numbers of neighbors (k).

from 9.44 m²/ha, using only undisturbed plots, to 12.37 m²/ha using all 1294 plots. For $k=9$, the values of the RSME went from 7.17 to 9.43. Further, in almost all cases, the producer's accuracy was less than reported when using only undisturbed plots. Including disturbed plots clearly introduced new sources of error. Since the exact date in which a disturbance occurred is unknown, a new temporal specification error is introduced; in particular, if the disturbance took place between or after the satellite image dates. For some of the disturbances (for example clearcutting and land clearing), the radiance values could change dramatically in a few months because of vegetation growth. Finally, the inclusion of disturbed plots incorporates new and extraordinary plot classes for which the pool of neighboring values is very small.

Finally, in the course of these trials, we identified one observation that appeared to have a serious error in the field

or plot location data. The location appeared to be young forest on the imagery and in the field plot classification, but basal area and volume data indicated a mature forest. Deleting that observation and repeating the analysis in Table 2a with $n=964$ plots reduced the crossvalidation estimate of RMSE from 9.44 to 8.87, a considerable reduction. It thus appears that the kNN technique can be quite sensitive to errors in data, especially with small values of k .

4.2. Cover type estimation

The analysis described in Section 4.1 was also developed for forest type classification. The cover type classes and codes used are shown in Table 1. In most cases, the prediction rule with the highest crossvalidation OA (i.e., smallest Err_{CV}) was considered the best. Nevertheless, the confusion matrix, Kappa estimator and producer's and user's accuracy were examined and considered in each trial.

4.2.1. Distance metric evaluation

The OA in classification comparing Mahalanobis and Euclidean distance is shown in Fig. 12. There is a rapid early increase in the accuracy of the technique as the number of neighbors is increased. However, as with volume and basal area estimation, the marginal increase in precision ultimately diminishes with a large number of neighbors. Note also that the value of the OA when $k=1$ and $k=2$ is the same. This is an artifact of the weighted mode estimator and the condition that in case of a tie the mode value is that of the closest neighbor. Again, the use of Mahalanobis distance did not improve the quality of the estimation. However, the maximum difference between Mahalanobis and Euclidean estimates of OA was always less than 1.5% for any value of k . Given these results, Euclidean distance was retained as the metric used in subsequent trials.

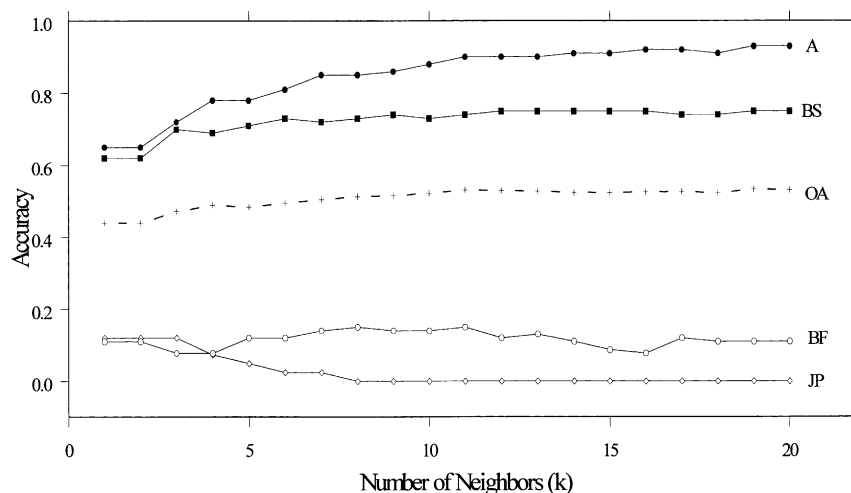


Fig. 18. Overall classification accuracy (OA) and producer's accuracy for some representative classes and different numbers of neighbors (k). Cover type symbols are A=Aspen, BS=Black Spruce, BF=Balsam Fir, and JP=Jack Pine.

4.2.2. Neighbor's weighting functions

Fig. 13 shows the results of applying different neighbor's weighting functions. Like basal area and volume estimation, there was no evidence supporting any of the weighting functions over the simple mean estimator. Thus, for subsequent trials, Eq. (2), with $t=0$, was used to allocate equal weights to neighbors.

4.2.3. Image filtering

Fig. 14 shows the results of the image filtering trials. Estimation was slightly more precise using the 4×3 pixel window median filtering when using one or two neighbors, while with three or more neighbors the three filtering options tested produced approximately the same result. The maximum gain in accuracy due to the use of a filter, when $k=1$, was approximately 4% (almost 10% relative to the estimation).

We expected a filter-related OA improvement closely related to classification improvement in a single class. In particular, we expected improvements showing a smoothing effect due to the use of filters. This was not the case. This differs with our previous experience applying kNN for continuous variables, and it may be related to the aggregated nature of a variable such as forest type.

4.2.4. Multitemporal image vs. single-date approach

Fig. 15 shows the results of these trials involving imagery dates. None of the single dates outperformed the use of the multitemporal approach. Using all 18 bands produced 5% more accuracy for $k=1$ or $k=2$, and 2% on average, than the closest competitor for $k>2$. For single dates, the summer image produced the second-best estimation. This is in contrast to results for basal area and volume estimation where a winter date performed best.

Table 5

Cover type classification confusion matrices for (a) $k=1$ and (b) $k=10$ (a) $k=1$, OA estimator; crossvalidation: 0.4402

		Reference data (FIA forest cover types)														
Confusion matrix		JP	RP	WP	BF	BS	NWC	T	WS	EAS	MBB	A	PB	BP	User's accuracy	
Classified	JP	5	1	1	2	10	3	1	0	1	2	15	3	2	0.11	
	RP	1	2	2	5	2	0	0	1	0	0	2	2	0	0.12	
	WP	1	2	1	0	0	1	0	0	1	0	3	0	0	0.11	
	BF	3	2	1	11	16	2	3	2	5	2	37	7	0	0.12	
	BS	11	2	0	20	126	11	22	4	3	0	19	3	1	0.57	
	NWC	1	2	0	5	9	7	1	1	2	0	0	1	0	0.22	
	T	1	1	0	5	16	3	10	0	1	1	14	1	0	0.19	
	WS	0	3	0	3	1	1	1	0	0	1	5	2	0	0	
	EAS	1	0	1	7	2	2	3	0	14	1	19	2	0	0.27	
	MBB	1	0	0	3	0	0	1	1	3	8	15	2	1	0.23	
	A	14	8	3	36	19	2	11	4	30	20	323	39	11	0.62	
	PB	1	1	0	6	1	3	1	1	5	4	33	16	2	0.22	
	BP	1	0	0	1	2	0	0	0	2	3	12	3	3	0.11	
Producer's accuracy		0.12	0.083	0.11	0.11	0.62	0.2	0.19	0	0.21	0.19	0.65	0.2	0.15		

(b) $k=10$, OA estimator; crossvalidation: 0.523

		Reference data													
Confusion matrix		JP	RP	WP	BF	BS	NWC	T	WS	EAS	MBB	A	PB	BP	User's accuracy
Classified	JP	0	1	0	0	1	1	0	0	0	0	1	0	0	0
	RP	0	0	0	2	1	0	0	1	0	0	1	0	0	0
	WP	0	0	1	0	0	1	0	0	0	0	0	0	0	0.5
	BF	3	4	1	15	13	7	3	3	4	1	20	3	0	0.19
	BS	11	5	0	25	149	10	26	2	2	2	21	6	0	0.58
	NWC	3	2	1	1	5	7	0	1	1	1	2	2	0	0.27
	T	0	0	0	0	1	0	2	0	0	0	2	0	0	0.4
	WS	0	0	0	1	0	0	0	0	0	0	0	0	0	0
	EAS	0	0	0	2	0	3	1	0	6	0	6	0	0	0.33
	MBB	0	0	0	0	1	0	0	0	1	2	2	0	0	0.33
	A	24	12	6	57	32	4	22	7	52	36	439	66	19	0.57
	PB	0	0	0	1	1	2	0	0	1	0	6	4	1	0.25
	BP	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Producer's accuracy		0	0	0.11	0.14	0.73	0.2	0.037	0	0.09	0.048	0.88	0.049	0	

General information: distance function = Euclidean, weighting function = equal, 1195 plots, 18-band multitemporal image, no band weighting parameters, and penalty = 0.

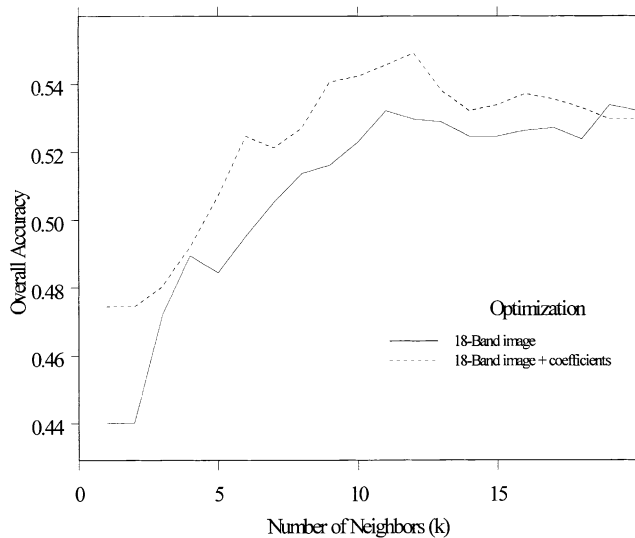


Fig. 19. OA of cover type classification for the 18-band image compared with the use of the same image with coefficients generated using the downhill simplex optimization method for different numbers of neighbors (k).

4.2.5. Image enhancement and ancillary information

The next step was to compare the performance of the 18-band image against image composites of tasseled cap indices and other VIs. Here, the indices did not provide better results than using the 18-band image (Fig. 16). Coefficients associated with the tasseled cap provided some further class discrimination, however, the OA improvement was very small in magnitude, associated only with the most populated classes, and to the detriment of the less populated classes.

As with basal area and volume estimation, using ancillary ECS information as the basis for a penalty value in the distance function did not help in the estimation. For any value of penalty tested, the OA results were the same. Again, this is probably due to the coarse resolution of the ancillary information that allowed for nearest neighbors to be consistently found within the same ECS class. In addition, it was quite possible for a cover type to appear in several if not all of the ECS classes.

Looking for an alternative source of ancillary information, tests were performed on the use of the variable physiographic class, as recorded on FIA plots (xeric, xeromesic, mesic, hydromesic, and hydric). While not a mapped variable, it is a reasonable surrogate for a soil moisture, drainage, or site quality variable. Using only FIA field plot records of this variable, but considering it as an ancillary source of information, we found it could easily boost the performance of the kNN 4–5% using the penalty value approach. Typically penalty values ranged from 1 to 25. The idea behind this test was to assess the utility of a currently unavailable, but potentially very useful (and feasible) map of localized growing conditions.

We also tested the incorporation of this new ancillary information as a feature in the distance computations. Appropriate weights for the ancillary variable were computed using the previously noted downhill simplex method and the results were remarkably similar to using this variable under the penalty system. The downhill simplex method identified the variable and scaled it appropriately. This result points to the possibility of incorporating several sources of ancillary information simultaneously.

Table 6

kNN classification summary report when $k=1$

$k=1$, Overall classification accuracy estimator

Kappa	0.3132														
Apparent	1														
Crossvalidation	0.4745														
Leave-one-out	0.4428														
bootstrap	0.5211														
Bootstrap 0.632 +															
Confusion matrix															
		Reference data													
		JP	RP	WP	BF	BS	NWC	T	WS	EAS	MBB	A	PB	BP	User's accuracy
Classified	JP	8	1	0	4	9	1	2	1	0	1	8	0	3	0.21
	RP	1	2	2	1	2	0	0	0	0	0	4	0	0	0.17
	WP	0	4	0	1	0	1	0	0	0	0	3	1	0	0
	BF	4	3	1	19	7	4	3	2	9	4	26	6	0	0.22
	BS	6	5	1	16	129	8	18	4	2	1	27	2	0	0.59
	NWC	0	0	1	2	6	8	2	1	2	0	3	3	0	0.29
	T	1	0	0	7	15	0	11	0	3	0	8	1	0	0.24
	WS	2	1	0	4	4	2	0	1	1	0	4	0	0	0.053
	EAS	1	0	0	8	1	4	2	0	19	3	14	4	2	0.33
	MBB	1	0	1	4	0	0	0	0	2	7	17	6	0	0.18
	A	15	7	3	30	28	4	14	5	21	15	339	38	6	0.65
	PB	1	1	0	6	2	3	1	0	4	9	40	20	5	0.22
	BP	1	0	0	2	1	0	1	0	4	2	7	0	4	0.18
Producer's accuracy		0.2	0.083	0	0.18	0.63	0.23	0.2	0.071	0.28	0.17	0.68	0.25	0.2	

General information: distance function = Euclidean, weighting function = equal, 1195 plots, 18-band multitemporal image, one neighbor ($k=1$), using band weighting parameters, 200 bootstrap samples and penalty=0.

Table 7

Photo classification vs. ground cover type of inventory plots from Deegan and Befort (1990) (Table 1)

Photo classification	Ground classification										Total
	JP	R/W	S–F	BS	WC	TK	LH	NH	A–B	N/O	
Jack Pine	30	6	9	8					12		65
Red/White Pine	3	6				1			7		17
White Spruce/Balsam Fir	7	5	150	83	67	10	27	3	86	3	441
Black Spruce	6	1	28	246	33	56	1		18	7	396
White Cedar				10	17		2		1	3	33
Tamarack	1	1	3	14	21	21				1	53
Lowland Hdwd	1		21	21	33	13	22	6	42	5	164
North. Hdwds			5		3		6	28	22		64
Aspen–Birch	11	36	191	65	38	10	117	105	1416	18	2007
Nonstock/Other	2		4	6	1	4			18		35
Total	61	55	411	453	204	115	175	142	1622	37	3275
No. of correct	30	6	150	246	17	21	22	28	1416	0	1936
Percent correct	0.492	0.109	0.365	0.543	0.083	0.183	0.126	0.197	0.873	0.00	0.591
KHAT											0.379
STDERR KHAT											0.013

4.2.6. Number of neighbors

The OA in kNN classifications using Euclidean distance, equal weighting among neighbors, a median 4×3 window median filter and the 18-band image is depicted in Fig. 17. The values for the OA increased approximately 5% (more than 10% relative to the estimation) when the number of neighbors was increased from one to five. After the number of neighbors reached 10, the marginal increase in precision was smaller than 1.0%.

Subsequently, we examined the behavior of the classification OA with respect to the confusion matrix producer's accuracy for some representative classes in Fig. 18. Although increasing k significantly increases the OA, there is a corresponding reduction in the producer's accuracy for the less populated classes. Bias occurs when these classes are pulled towards or misclassified into the most populated classes.

In particular, if we consider the less populated classes (exemplified by Jack Pine, JP in Fig. 18) increasing the number of neighbors from one to eight causes a 100% reduction (from 12% to 0% producer's accuracy) in the accuracy for classifying Jack Pine plots. A similar situation occurs for almost all classes. The large Aspen and Black Spruce classes were the only exceptions. The improvement in the OA for the classification when increasing k is due to improvements in the most populated classes. These most populated classes tend to assimilate the other classes. The only exception was Balsam Fir (a medium populated class) for which the producer's accuracy remained almost constant.

These results are also illustrated in Table 5 showing producer's accuracy and user's accuracy for cover type when $k=1$ and $k=10$. This result implies that increasing the number of neighbors produces an undesirable reduction of variability, especially if the goal of using kNN is map production. This problem is reduced by using fewer neighbors and disappears using only one neighbor.

The results show a trade-off between accuracy and the objectives pursued using the kNN method. Clearly, the

decision about how many neighbors to use again depends on the objective of the estimation. Using only OA, the conclusion reached would have been to use $k=10$ neighbors to construct a cover type map. However, for map production, $k=1$ may be preferred since it retains the full range of cover types present in the data. This choice of k becomes more important as the number of classes increases. Here, we have chosen to proceed with $k=1$.

4.2.7. Weighting parameters

Given the choice of $k=1$ to a cover type map, the next step was to compute coefficients (weighting parameters) for each TM band. This step was accomplished using $k=1$ and the same downhill simplex optimization method employed in basal area and volume estimation.

The coefficients shown in Table 3 were applied in the context described in Eq. (5) to compute kNN estimates. Fig.

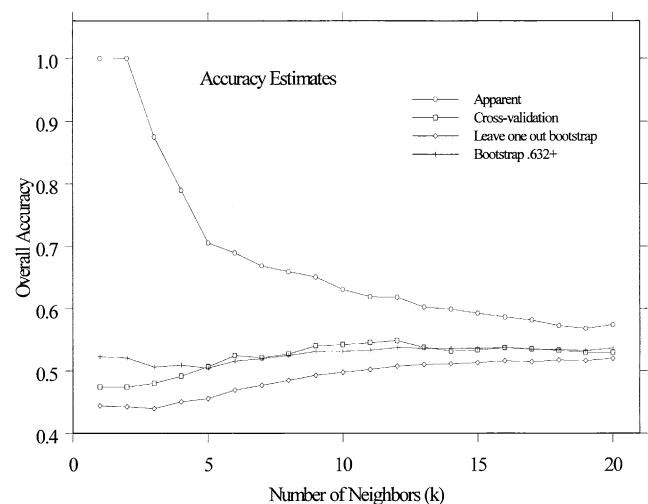


Fig. 20. OA estimation using different statistical procedures for different numbers of neighbors (k) and 50 bootstrap samples.

Table 8

kNN summary report for a classification system considering three classes (softwoods, hardwoods, and mixed hardwoods–softwoods) with $k=1$

$k=1$, Overall classification accuracy estimator				
Crossvalidation:	0.6402			
Kappa:	0.4519			
	Reference data			
Confusion matrix	Softwoods	Hardwoods	Mixed	User's accuracy
Softwoods	214	15	79	0.69
Hardwoods	23	268	127	0.64
Mixed	71	115	283	0.60
Producer's accuracy	0.69	0.67	0.58	

General information: distance function = Euclidean, weighting function = equal, 1195 plots, 18-band multitemporal image, using band weighting parameters and penalty = 0.

19 compares the performance of the kNN estimator with and without these weighting parameters. Gains with optimization were modest except for $k=1$ where the OA increased almost 4%.

4.2.8. Overall accuracy estimation

Table 6 shows the summary report of the results obtained applying the kNN method for forest cover type when $k=1$, using weighting parameters and 200 bootstrap samples. The crossvalidation estimator of OA is equal to 0.47 and the bootstrap 0.632+ estimator of OA is 0.52. These results may seem poor, but they are fairly representative of the type of results frequently obtained in this region. With many cover types and very little topographic variation, it is a considerable test. Deegan and Befort (1990) analyzed the accuracy levels achieved by FIA classification procedures using aerial photography. Table 7 shows their results comparing photo vs. ground cover type classifications for the same general region and type of forest in 1977. The OA in the classification obtained through the use of aerial photographs was 0.59. Our confusion matrix (Table 6) and theirs shares the same structure in which most of the omission errors are in the most populated classes (Aspen–Birch and Black Spruce).

An additional factor to consider is that the FIA plots were not preselected for training. Instead, they represent an essentially random sample of points in the forest. Straddler plots (those that cross cover types and various stand conditions) were included. Thus, the characterization of accuracy here is deemed more informative, statistically, than results from studies where training sites are developed in a non random manner and with sample sizes by training class that are not in proportion to population frequencies. Lacking common procedure for site selection and random sample, studies are often difficult to compare.

4.2.9. Prediction error estimation

The objective of seeking improvements in error estimation through bootstrapping techniques was to obtain an estimate of true error as good as that obtained using cross-validation, but with smaller variance. Fig. 20 shows the behavior of the different estimation procedures for different

number of neighbors and 50 bootstrap samples. Note that the Bootstrap 0.632+ is a nearly unbiased estimator with smaller variability than crossvalidation. It works fine with overfitted rules such as nearest neighbors. Following Efron and Tibshirani's (1997) reasoning for computing the standard error of $\text{Err}_{0.632+} (1 - \text{OA})$, we multiplied the Err_{LOOB} empirical confidence interval limits by $\text{Err}_{0.632+} / \text{Err}_{\text{LOOB}}$ to get a confidence interval of the 0.632+ bootstrap. Therefore, (0.50, 0.55) constitutes a 95% confidence interval for bootstrap 0.632+ OA.

4.2.10. Map construction

The experience generated from the above trials was transferred to the construction of thematic maps. Using the kNN method with an 18-band image composite, Euclidean distance and the nearest neighbor, a cover type map (a portion of the map is shown in Fig. 21) was generated for the forest area in St. Louis County. The classes considered in this thematic map correspond to those in Table 1. A forest–nonforest mask was used to separate lakes and other nonforest lands in this map.

Table 9

Distribution of cover types in St. Louis County as determined from different sources

Code	Cover type class	Percentage of FIA plots/ forest area	Percentage of classified in test sample	Percentage of image
JP	Jack Pine	3.4	3.18	3.11
RP	Red Pine	2.0	1.00	1.95
WP	White Pine	0.8	0.84	0.75
BF	Balsam Fir	8.7	7.36	9.32
BS	Black Spruce	17.1	18.33	18.69
NWC	Northern White Cedar	2.9	2.34	3.62
T	Tamarack	4.5	3.85	5.06
WS	White Spruce	1.2	1.59	1.00
EAS	Elm–Ash–Soft Maple Group	5.6	4.85	4.75
MBB	Maple–Birch–Beech Group	3.5	3.18	3.03
A	Aspen	41.8	43.93	39.90
PB	Paper Birch	6.8	7.70	6.84
BP	Balsam Poplar	1.7	1.84	1.64

Although the error evaluation results may seem relatively poor the map results are very encouraging. Table 8 shows the cover type proportions as determined by different sources. The map production effort was very successful (better than the test sample) in reproducing the FIA's plot cover type percentage in the area. This suggests the method has better matching capabilities than those indicated in the accuracy assessment.

4.2.11. The FIA classification approach

The algorithm-based classification system applied by FIA (Hansen & Hahn, 1992) is an important factor in the precision of the results obtained. One of the most limiting factors in the classification is the lack of a mixed hardwoods–softwoods forest type. Trying to check the performance of the kNN method further, we applied a simple basal area classification to form a three-class scheme, hardwoods, conifers and mixed hardwoods–conifers. The pure hardwoods and pure conifers classes contained less than 20% of the other class. The results of the kNN estimation focused on these three classes are shown in Table 9 and Fig. 22. We obtained an OA of 0.64 when applying the crossvalidation

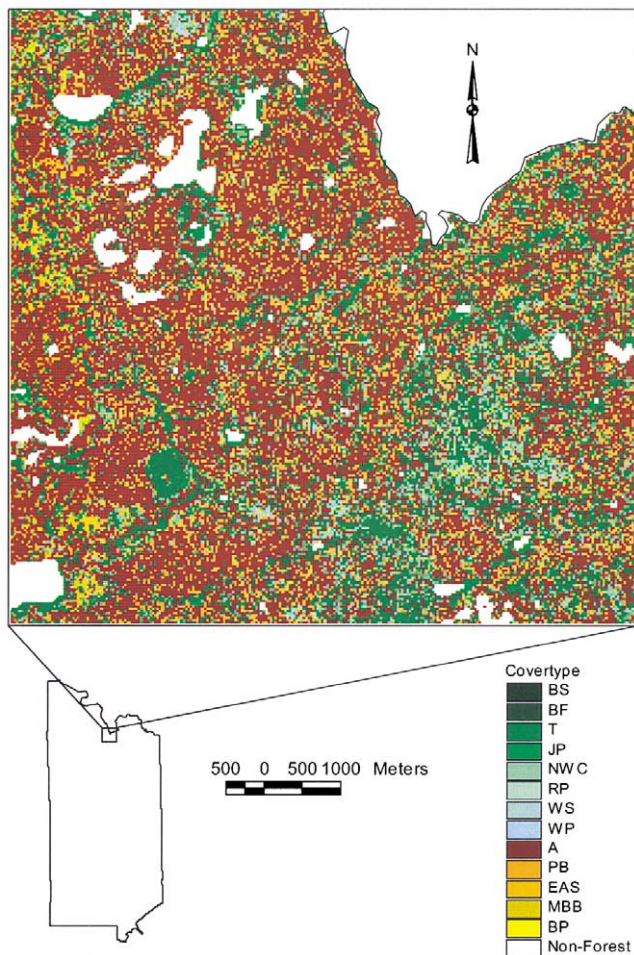


Fig. 21. KNN classification map of FIA based forest cover types for a portion of St. Louis County, MN.

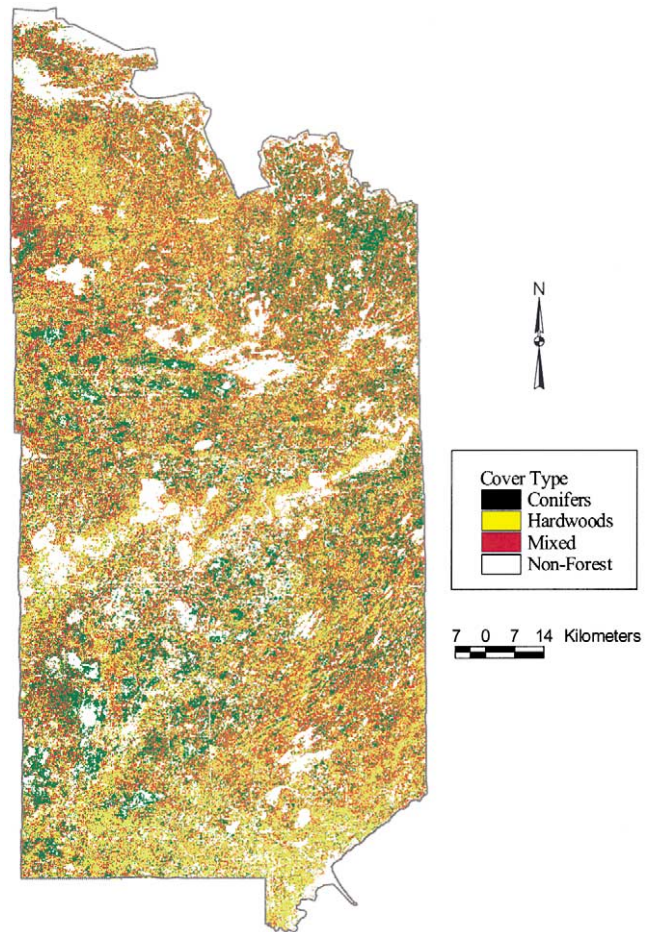


Fig. 22. KNN classification map of FIA based conifers, hardwoods, and mixed species cover types for St. Louis County, MN.

estimator. These results ($OA=0.64$) suggest a much improved OA is possible if mixed and other similar classes are combined.

5. Conclusions

The kNN method is very promising for propagating forest stand density, volume, and cover type through the landscape. The simplicity of this method and its role in post stratification provides a very feasible tool for wall-to-wall mapping of forest variables and local to landscape scale estimation. However, there is room for improvement.

There are several important advantages of the method. For years, analysts have been looking for techniques to combine inventory sources. The kNN method is a versatile technique with potential for combining different sources of information, not only from outside of a region of interest, but even from different forest inventory designs. The combination of different remote sensors is straightforward since the method is based solely in the search for similar units. In contrast to conventional image classification where

classes are interpreted considering only the plots included in them, here, the classes are defined by the ground data in a poststratification approach.

The method is also easy to understand while allowing flexibility in the incorporation of ancillary information. Another advantage is that a number of forest variables can be estimated at the same time. Since the same neighbors are involved, the quality of the estimation relies on the correlation among the variables and the image features.

The number of nearest neighbors to employ in an estimation problem is determined by the particular goals of a survey. When applying kNN for map production and using only the nearest neighbor, the estimator is unbiased and the range in variability of the sample is largely preserved. Using only one neighbor in estimation procedures also opens the door to techniques such as imputation for error description. Using more than one neighbor is appropriate to produce estimates of forest variables over large areas.

There are several research needs. Since location in the field is the main relational feature of the kNN technique, priority should be given to understanding the need for plot location accuracy. Additionally, here, we followed a multi-temporal approach, yet, multisensor methods could further improve the quality of estimation. Further, it is appropriate to examine how kNN methods may best be used in forest inventory, for stratification for variance reduction, for stratum size estimation, and for local estimation. Research is also needed to assess the best ways to use ancillary information. Identifying ancillary factors driving local response would be very helpful to improve the kNN estimation procedure.

The quality of cover type classification results is comparable with the quality obtained using parametric approaches, but with many advantages. Among its advantages, the kNN method is a very “transparent” technique. Since it is based solely on the search for similar units, it is easy to understand. The key to success is having enough ground samples to cover all variations in tree size and stand density for each cover type. Finally, since location in the field is the main relational feature of the kNN technique, priority should be given to obtaining precise location information.

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